30 Days SQL Micro Course

Practise Sql Queries 1

Queries using only:

- CREATE TABLE
- INSERT
- SELECT
- WHERE
- ORDER BY
- LIMIT
- UPDATE
- DELETE

Practise link : https://dbfiddle.uk/vKsjhDAw

Step 1: CREATE TABLE

Question

Create a table to store employee information.

Think

What information do we need?

Column	Data Type
EmployeeID	INT
Name	VARCHAR(50)
Department	VARCHAR(30)
Salary	INT
Age	INT

Query

```
CREATE TABLE Employees (
   EmployeeID INT,
   Name VARCHAR(50),
   Department VARCHAR(30),
   Salary INT,
   Age INT
);
```

What happened?

A table named Employees is created.

Step 2: INSERT DATA

Question

Add employee records.

Query

```
INSERT INTO Employees
VALUES (1, 'Amit', 'IT', 50000, 25);
INSERT INTO Employees
VALUES (2, 'Neha', 'HR', 60000, 28);
INSERT INTO Employees
VALUES (3, 'Rahul', 'IT', 70000, 30);
```

INSERT INTO Employees

VALUES (4, 'Priya', 'Sales', 45000, 24);

INSERT INTO Employees

VALUES (5, 'Ravi', 'HR', 55000, 29);

```
INSERT INTO Employees
VALUES (1, 'Amit', 'IT', 50000, 25);

INSERT INTO Employees
VALUES (2, 'Neha', 'HR', 60000, 28);

INSERT INTO Employees
VALUES (3, 'Rahul', 'IT', 70000, 30);

INSERT INTO Employees
VALUES (4, 'Priya', 'Sales', 45000, 24);

INSERT INTO Employees
VALUES (5, 'Ravi', 'HR', 55000, 29);
```

```
INSERT 0 1
INSERT 0 1
INSERT 0 1
INSERT 0 1
INSERT 0 1
```

```
SELECT *
FROM Employees;
```

employeeid	name	department	salary	age
1	Amit	IT	50000	25
2	Neha	HR	60000	28
3	Rahul	IT	70000	30
4	Priya	Sales	45000	24
5	Ravi	HR	55000	29

SELECT 5

Step 3: SELECT

Question

Show all employees.

Think

Need:

- **All columns**

Query

SELECT *
FROM Employees;

```
SELECT *
FROM Employees;
```

employeeid	name	department	salary	age
1	Amit	IT	50000	25
2	Neha	HR	60000	28
3	Rahul	IT	70000	30
4	Priya	Sales	45000	24
5	Ravi	HR	55000	29

SELECT 5

Step 4: SELECT Specific Columns

Question

Show employee names only.

Think

Need:

- **Name column**

Query

SELECT Name
FROM Employees;

Question

Show Name and Salary.

Query

SELECT Name, Salary
FROM Employees;

Step 5: WHERE

Question

Show employees older than 25.

Think

Filter Age.

SELECT *
FROM Employees
WHERE Age > 25;

Question

Show HR employees.

**SELECT *
FROM Employees
WHERE Department = 'HR';**

Question

Show employees with salary greater than 50000.

**SELECT *
FROM Employees
WHERE Salary > 50000;**

```
SELECT *
FROM Employees
WHERE Age > 25;
```

employeeid	name	department	salary	age
2	Neha	HR	60000	28
3	Rahul	IT	70000	30
5	Ravi	HR	55000	29

SELECT 3

```
SELECT *
FROM Employees
WHERE Department = 'HR';
```

employeeid	name	department	salary	age
2	Neha	HR	60000	28
5	Ravi	HR	55000	29

SELECT 2

```
SELECT *
FROM Employees
WHERE Salary > 50000;
```

employeeid	name	department	salary	age
2	Neha	HR	60000	28
3	Rahul	IT	70000	30
5	Ravi	HR	55000	29

SELECT 3

Step 6: ORDER BY

Question

Show highest salary first.

**SELECT *
FROM Employees
ORDER BY Salary DESC;**

Question

Show lowest salary first.

**SELECT *
FROM Employees
ORDER BY Salary;**

```
SELECT *
FROM Employees
ORDER BY Salary DESC;
```

SELECT 3

employeeid	name	department	salary	age
3	Rahul	IT	70000	30
2	Neha	HR	60000	28
5	Ravi	HR	55000	29
1	Amit	IT	50000	25
4	Priya	Sales	45000	24

SELECT 5

```
SELECT *
FROM Employees
ORDER BY Salary;
```

employeeid	name	department	salary	age
4	Priya	Sales	45000	24
1	Amit	IT	50000	25
5	Ravi	HR	55000	29
2	Neha	HR	60000	28
3	Rahul	IT	70000	30

SELECT 5

Step 7: LIMIT

Question

Show first 3 employees.

SELECT *
FROM Employees
LIMIT 3;

SELECT 5

```
SELECT *
FROM Employees
LIMIT 3;
```

employeeid	name	department	salary	age
1	Amit	IT	50000	25
2	Neha	HR	60000	28
3	Rahul	IT	70000	30

SELECT 3

Step 8: UPDATE

Question

Change Amit's salary to 55000.

UPDATE Employees
SET Salary = 55000
WHERE Name = 'Amit';

Check:

SELECT *
FROM Employees;

```
UPDATE Employees
SET Salary = 55000
WHERE Name = 'Amit';
```

UPDATE 1

```
SELECT *
FROM Employees;
```

employeeid	name	department	salary	age
2	Neha	HR	60000	28
3	Rahul	IT	70000	30
4	Priya	Sales	45000	24
5	Ravi	HR	55000	29
1	Amit	IT	55000	25

SELECT 5

Step 9: DELETE

Question

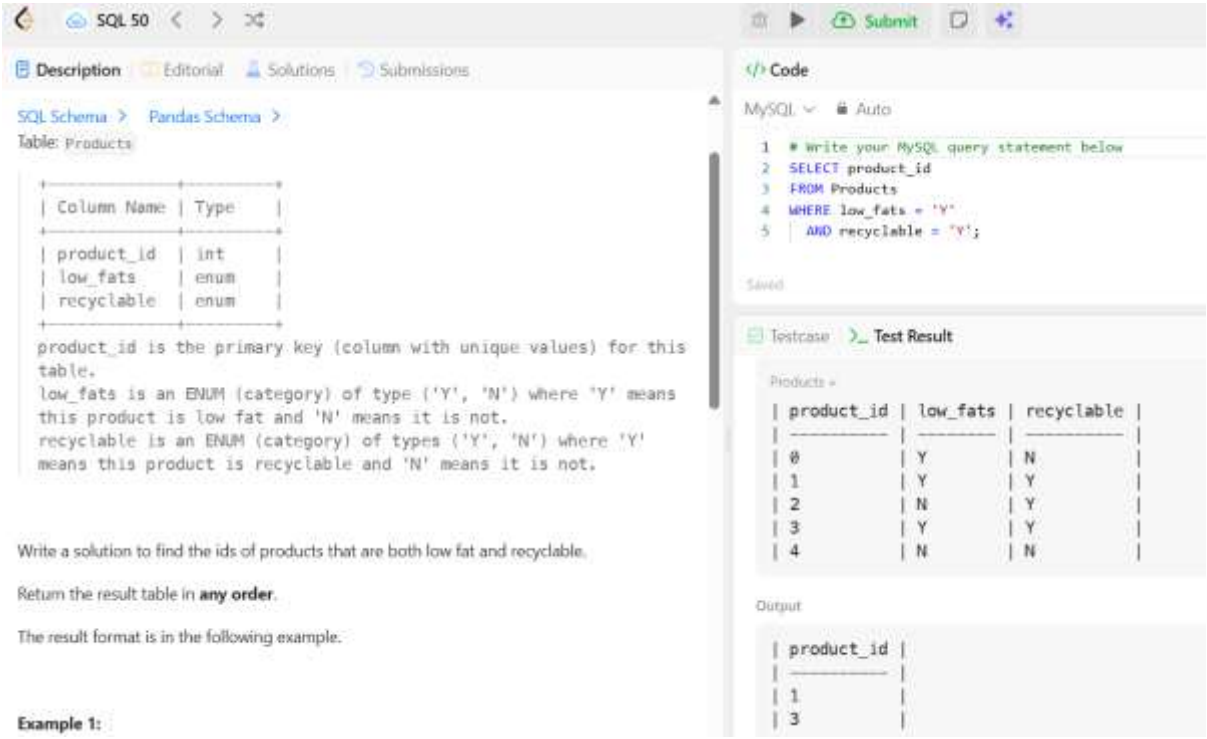
Delete Priya's record.

DELETE FROM Employees
WHERE Name = 'Priya';

Check:

SELECT *
FROM Employees;

Solving sql queries: https://leetcode.com/problems/recyclable-and-low-fat-products/?envType=study-plan-v2&envId=top-sql-50



filter by specific product_ids. Instead, you need to select products where:

- **low_fats = 'Y'**
- **recyclable = 'Y'**

Description | Editorial | Solutions | Submissions

584. Find Customer Referee

Solved

Easy | Topics | Companies | Hint

SQL Schema > Pandas Schema >

Table: Customer

Column Name	Type
id	int
name	varchar
referee_id	int

In SQL, id is the primary key column for this table.
Each row of this table indicates the id of a customer, their name, and the id of the customer who referred them.

Find the names of the customer that are either:

1. **referred by** any customer with `id != 2`.
2. **not referred by** any customer.

Return the result table in **any order**.

The result format is in the following example.

Code

MySQL | Auto

```
# Write your MySQL query statement below
SELECT name
FROM Customer
WHERE referee_id != 2
   OR referee_id IS NULL;
```

Saved

Testcase | Test Result

id	name	referee_id
1	Will	null
2	Jane	null
3	Alex	2
4	Bill	null
5	Zack	1
6	Mark	2

Output

name
Will
Jane
Bill
Zack

Include customers whose referee is NOT 2

Also include customers with no referee

SQL 50 | Submit

Description | Editorial | Solutions | Submissions

SQL Schema > Pandas Schema >

Table: world

Column Name	Type
name	varchar
continent	varchar
area	int
population	int
gdp	bigint

name is the primary key (column with unique values) for this table.
Each row of this table gives information about the name of a country, the continent to which it belongs, its area, the population, and its GDP value.

A country is **big** if:

- it has an area of at least three million (i.e., 3000000 km²), or
- it has a population of at least twenty-five million (i.e., 25000000).

Write a solution to find the name, population, and area of the **big countries**.

Return the result table in **any order**.

The result format is in the following example.

Code

MySQL | Auto

```
# Write your MySQL query statement below
SELECT name,population,area
FROM world
WHERE area >= 3000000
OR population >= 25000000;
```

Saved

Testcase | Test Result

name	continent	area	population	gdp
Afghanistan	Asia	652230	25500100	20343000000
Albania	Europe	28748	2831741	12960000000
Algeria	Africa	2381741	37100000	188681000000
Andorra	Europe	468	78115	3712000000
Angola	Africa	1246700	20609294	100990000000

Output

name	population	area
Afghanistan	25500100	652230
Algeria	37100000	2381741

Expected

| name | population | area |

- **at least 3,000,000 → >= 3000000**
- **more than 3,000,000 → > 3000000**
- **at most 3,000,000 → <= 3000000**

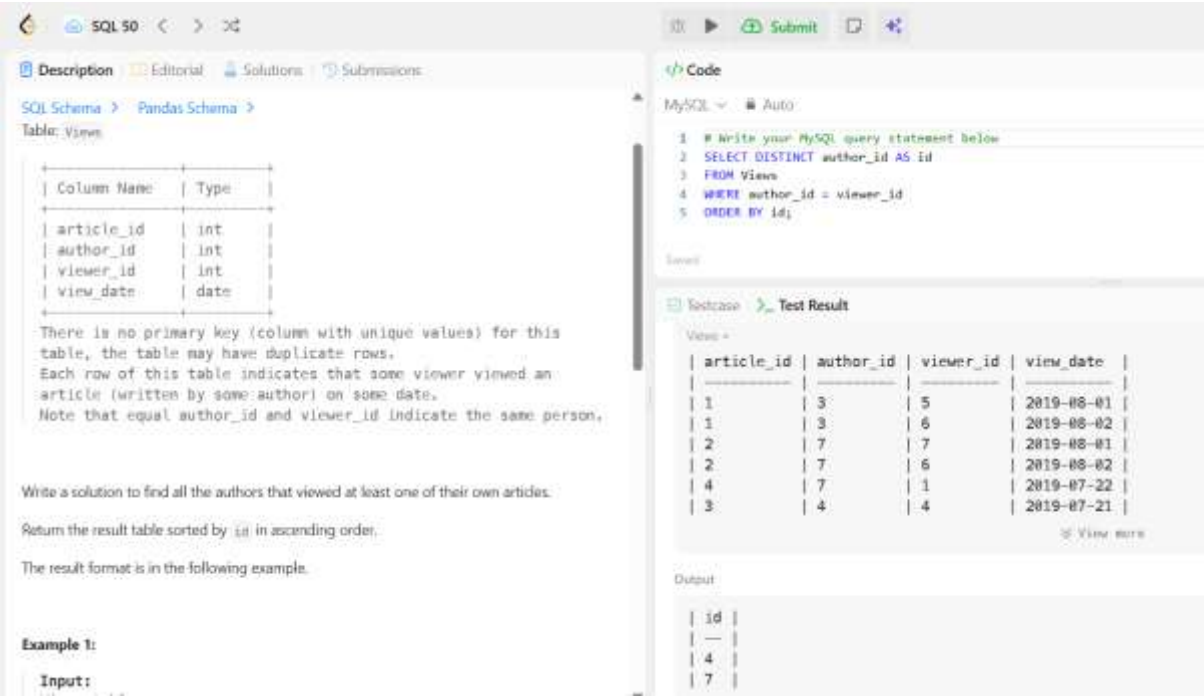
- **less than 3,000,000 → < 3000000**

A country is big if:

- **it has an area of at least 3,000,000**
- **OR it has a population of at least 25,000,000**

Translate directly to SQL:

area >= 3000000
OR population >= 25000000



equal author_id and viewer_id indicate the same person

and

find all the authors that viewed at least one of their own articles

Translate to SQL logic

If an author viewed their own article:

author_id = viewer_id

From the sample:

article_id	author_id	viewer_id
2	7	7
3	4	4

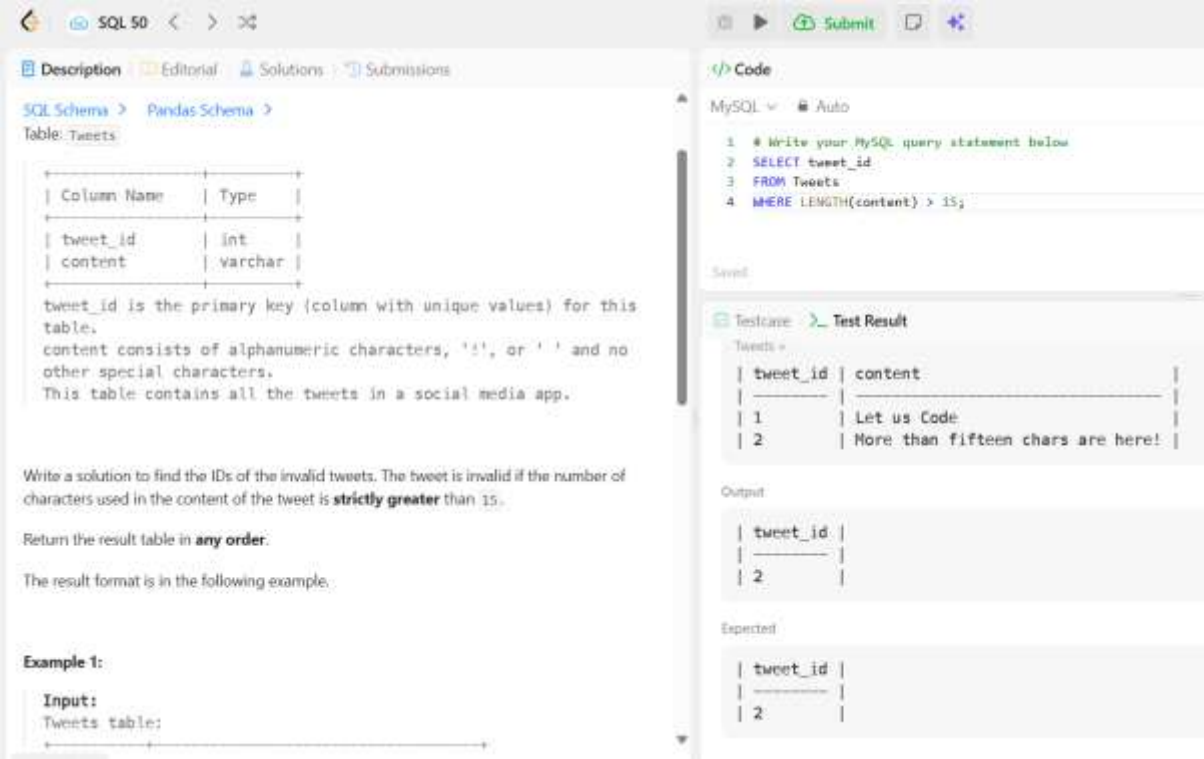
article_id author_id viewer_id

3 4 4

Authors 7 and 4 viewed their own articles.

Requirements

1. **Return column name as id**
2. **Remove duplicates (DISTINCT)**
3. **Sort ascending (ORDER BY)**



number of characters used in the content is strictly greater than 15

Translate English → SQL

- **"strictly greater than 15" → > 15**
- **Count characters in a string → LENGTH()**
- **Useful SQL String Functions**

Function	Purpose
`LENGTH(col)`	Number of characters
`UPPER(col)`	Uppercase text
`LOWER(col)`	Lowercase text
`CONCAT(a,b)`	Join strings

Common Questions → SQL

Need 2 conditions?

WHERE A AND B

Example:

WHERE low_fats='Y' AND recyclable='Y'

Need either condition?

WHERE A OR B

Example:

WHERE area >= 3000000
OR population >= 25000000

Need empty values?

WHERE column IS NULL

Remove duplicate rows?

SELECT DISTINCT column
FROM table;

Rename a column?

SELECT author_id AS id
FROM Views;

Sort data?

ORDER BY id

Count letters in text?

LENGTH(content)